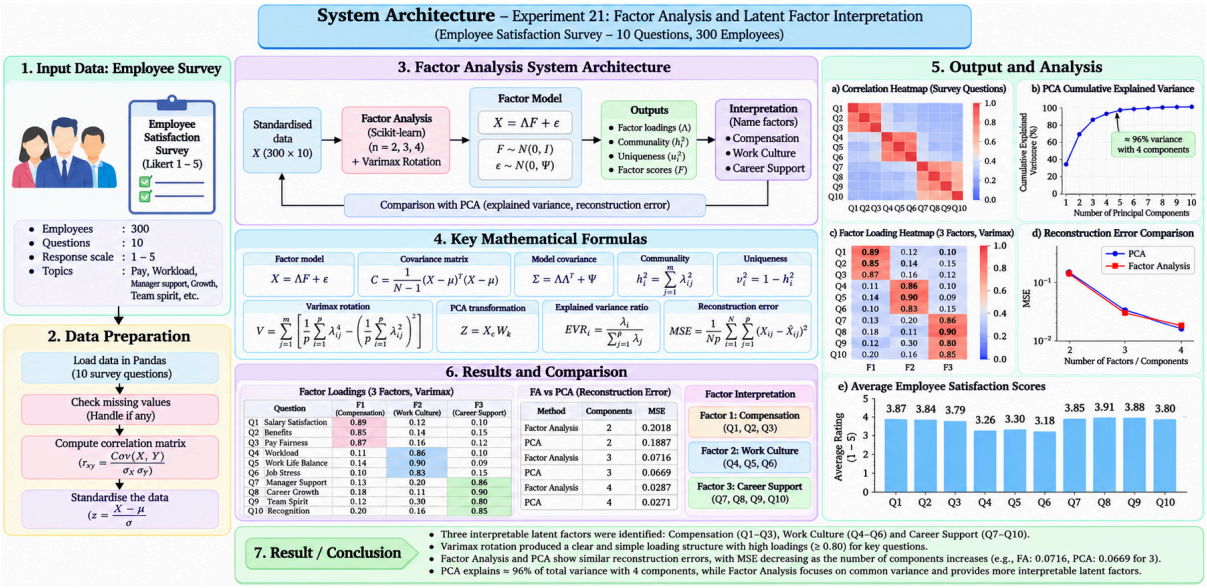
# Experiment 21: Factor Analysis and Latent Factor Interpretation

## 1. Aim

To perform Factor Analysis on an employee-satisfaction survey containing responses from 300 employees across 10 questions. To identify hidden factors influencing employee satisfaction, examine correlations between survey questions, apply Factor Analysis using 2, 3, and 4 factors, perform Varimax rotation, visualize factor loadings using heatmaps, interpret the latent factors, and compare Factor Analysis with Principal Component Analysis (PCA) using Scikit-learn, NumPy, Pandas, and Matplotlib.



## 2. Mathematical Formulas

### 1. Factor Analysis Model

$$\boxed{X = \Lambda F + \epsilon}$$

Where:

- $X$ = Observed survey responses.
- $\Lambda$ = Factor loading matrix.
- $F$ = Latent factors.
- $\epsilon$ = Unique error.

### 2. Covariance Matrix

$$\boxed{C = \frac{1}{N-1}(X - \mu)^T (X - \mu)}$$

### 3. Correlation Coefficient

$$\boxed{r_{xy} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}}$$

### 4. Factor Analysis Covariance Model

$$\boxed{\Sigma = \Lambda \Lambda^T + \Psi}$$

Where $\Psi$ is the diagonal matrix of unique variances.

### 5. Communality

$$\boxed{h_i^2 = \sum_{j=1}^{m} \lambda_{ij}^2}$$

Communality represents the variance in a variable explained by the common factors, assuming standardized variables and orthogonal unit-variance factors.

### 6. Uniqueness

$$\boxed{u_i^2 = 1 - h_i^2}$$

### 7. Varimax Rotation

$$\boxed{V = \sum_{j=1}^{m} \left[ \frac{1}{p} \sum_{i=1}^{p} \lambda_{ij}^4 - \left( \frac{1}{p} \sum_{i=1}^{p} \lambda_{ij}^2 \right)^2 \right]}$$

Varimax rotation makes factor loadings easier to interpret by encouraging a simpler loading pattern.

### 8. PCA Transformation

$$\boxed{Z = X_c W_k}$$

**CODE**

```
# Experiment 21: Factor Analysis and Latent Factor Interpretation
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.decomposition import FactorAnalysis, PCA
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error
np.random.seed(42)
# ========================================
# 1. GENERATE EMPLOYEE SURVEY DATA
# ========================================
n_employees = 300
questions = [
    "Salary Satisfaction",
    "Benefits",
    "Pay Fairness",
    "Workload",
    "Work Life Balance",
    "Job Stress",
    "Manager Support",
    "Career Growth",
    "Team Spirit",
    "Recognition"
]
# Three underlying latent factors
# Compensation, Work Culture, Career Support
latent_factors = np.random.normal(
    size=(n_employees, 3)
)
# Factor loading structure
true_loadings = np.array([
    [0.90, 0.10, 0.10],
    [0.85, 0.10, 0.15],
    [0.88, 0.15, 0.10],
    [0.10, 0.85, 0.10],
    [0.10, 0.90, 0.10],
    [0.10, 0.82, 0.15],
    [0.10, 0.20, 0.85],
    [0.15, 0.10, 0.90],
    [0.10, 0.30, 0.80],
    [0.20, 0.15, 0.85]
])

# Add random survey variation
noise = np.random.normal(
    0, 0.35, (n_employees, 10)
)
responses = (
    latent_factors @ true_loadings.T
    + noise
)
# Convert into Likert-scale responses (1-5)
responses = np.clip(
    np.rint(3 + responses),
    1, 5
).astype(int)
```

```
df = pd.DataFrame(
    responses,
    columns=questions
)
print("EMPLOYEE SATISFACTION DATASET")
print(df.head())
print("\nDataset Shape:", df.shape)
print("\nMissing Values:", df.isnull().sum().sum())
# ========================================
# 2. EXPLORE SURVEY DATA
# ========================================
print("\nDESCRIPTIVE STATISTICS")
print(df.describe().round(2))
print("\nAVERAGE SATISFACTION SCORES")
print(df.mean().round(2))
plt.figure(figsize=(11, 5))
plt.bar(
    df.columns,
    df.mean()
)
plt.xticks(rotation=65, ha="right")
plt.ylabel("Average Rating")
plt.title("Average Employee Satisfaction Scores")
plt.ylim(0, 5)
plt.tight_layout()
plt.show()
# ========================================
# 3. CORRELATION MATRIX
# ========================================
correlation = df.corr()
print("\nCORRELATION MATRIX")
print(correlation.round(2))
plt.figure(figsize=(10, 8))
plt.imshow(
    correlation,
    cmap="coolwarm",
    vmin=-1,
    vmax=1
)
plt.colorbar(label="Correlation")
plt.xticks(
    range(10),
    questions,
    rotation=90
)
plt.yticks(range(10), questions)
plt.title("Employee Survey Correlation Heatmap")
plt.tight_layout()
plt.show()
# ========================================
# 4. STANDARDIZE THE DATA
# ========================================
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df)
print("\nStandardized Data Shape:", X_scaled.shape)
# ========================================
# 5. FACTOR ANALYSIS WITH VARIMAX
# ========================================
```

```
factor_counts = [2, 3, 4]
fa_models = {}
fa_loadings = {}
comparison_results = []
for n in factor_counts:
    fa = FactorAnalysis(
        n_components=n,
        rotation="varimax",
        random_state=42,
        max_iter=1000
    )
    scores = fa.fit_transform(X_scaled)
    # Shape: questions x factors
    loadings = fa.components_.T
    fa_models[n] = fa
    fa_loadings[n] = loadings
    # Approximate reconstruction
    reconstructed = (
        scores @ fa.components_
        + fa.mean_
    )
    mse = mean_squared_error(
        X_scaled,
        reconstructed
    )
    comparison_results.append({
        "Method": "Factor Analysis",
        "Components": n,
        "Reconstruction MSE": mse
    })
    print(f"\nFA WITH {n} FACTORS")
    print("Reconstruction MSE:", round(mse, 4))
# ========================================
# 6. DISPLAY FACTOR LOADING HEATMAPS
# ========================================
fig, axes = plt.subplots(
    1, 3,
    figsize=(17, 7)
)
for ax, n in zip(axes, factor_counts):
    loadings = fa_loadings[n]
    im = ax.imshow(
        loadings,
        cmap="coolwarm",
        vmin=-1,
        vmax=1,
        aspect="auto"
            ax.text(
                j, i,
                f"{loadings[i, j]:.2f}",
                ha="center",
                va="center",
                fontsize=8
            )
fig.colorbar(
    im,
    ax=axes,
    shrink=0.65,
```

```
    label="Factor Loading"
)
plt.suptitle(
    "Varimax Rotated Factor Loading Heatmaps"
plt.show()
# ========================================
# 7. INTERPRET THREE-FACTOR SOLUTION
# ========================================
loadings_3 = fa_loadings[3]
loading_df = pd.DataFrame(
    loadings_3,
    index=questions,
    columns=[
        "Factor 1",
        "Factor 2",
        "Factor 3"
    ]
)
print("\nTHREE-FACTOR LOADING MATRIX")
print(loading_df.round(3))
print("\nHIGH LOADING QUESTIONS")
for j in range(3):
    factor_name = f"Factor {j+1}"
    top_questions = (
        loading_df[factor_name]
        .abs()
        .sort_values(ascending=False)
        .head(3)
        .index.tolist()
    )

    print(f"\n{factor_name}:")
    for question in top_questions:
        print("-", question)
# ========================================
# 8. AUTOMATIC FACTOR NAMING
# ========================================
groups = {
    "Compensation": questions[0:3],
    "Work Culture": questions[3:6],
    "Career Support": questions[6:10]
}
factor_names = {}
for j in range(3):
    scores = {}

    for group_name, group_questions in groups.items():

        scores[group_name] = (
            loading_df.loc[
                group_questions,
                f"Factor {j+1}"
            ].abs().mean()
        )

    factor_names[f"Factor {j+1}"] = max(
        scores,
        key=scores.get
```

```
    )
print("\nINTERPRETED LATENT FACTORS")
for factor, name in factor_names.items():
    print(factor, ":", name)

# ========================================
# 9. COMMUNALITY AND UNIQUENESS
# ========================================
communalities = np.sum(
    loadings_3 ** 2,
    axis=1
)
uniqueness = fa_models[3].noise_variance_

factor_details = pd.DataFrame({
    "Question": questions,
    "Communality": communalities,
    "Uniqueness": uniqueness
})

print("\nCOMMUNALITY AND UNIQUENESS")
print(factor_details.round(3))
# ========================================
# 10. PCA COMPARISON
# ========================================
for n in factor_counts:

    pca = PCA(
        n_components=n,
        random_state=42
    )
    transformed = pca.fit_transform(
        X_scaled
    )
    reconstructed = pca.inverse_transform(
        transformed
    )
    mse = mean_squared_error(
        X_scaled,
        reconstructed
    )
    comparison_results.append({
        "Method": "PCA",
        "Components": n,
        "Reconstruction MSE": mse
    })
    print(f"\nPCA WITH {n} COMPONENTS")
    print("Explained Variance:",
          round(
              np.sum(pca.explained_variance_ratio_) * 100,
              2
          ), "%")
    print("Reconstruction MSE:", round(mse, 4))
# ========================================
# 11. FINAL COMPARISON TABLE
# ========================================

comparison_df = pd.DataFrame(
```

```
    comparison_results
)
print("\nFA VS PCA COMPARISON")
print(
    comparison_df
    .sort_values(["Components", "Method"])
    .round(4)
    .to_string(index=False)
)
# ========================================
# 12. RECONSTRUCTION ERROR GRAPH
# ========================================
plt.figure(figsize=(9, 5))
for method in ["Factor Analysis", "PCA"]:
    subset = comparison_df[
        comparison_df["Method"] == method
    ]
    plt.plot(
        subset["Components"],
        subset["Reconstruction MSE"],
        marker="o",
        label=method
    )
plt.xlabel("Number of Factors / Components")
plt.ylabel("Reconstruction MSE")
plt.title("Factor Analysis vs PCA Reconstruction Error")
plt.xticks([2, 3, 4])
plt.legend()
plt.grid(alpha=0.3)
plt.show()
# ========================================
# 13. PCA EXPLAINED VARIANCE
# ========================================
pca_full = PCA()
pca_full.fit(X_scaled)
plt.figure(figsize=(9, 5))
plt.plot(
    range(1, 11),
    np.cumsum(
        pca_full.explained_variance_ratio_
    ) * 100,
    marker="o"
)
plt.xlabel("Number of Principal Components")
plt.ylabel("Cumulative Explained Variance (%)")
plt.title("PCA Cumulative Explained Variance")
plt.xticks(range(1, 11))
plt.grid(alpha=0.3)
plt.show()
# ========================================
# 14. FINAL RESULT
# ========================================
print("\nEXPERIMENT 21 COMPLETED SUCCESSFULLY")
print("Employees:", n_employees)
print("Survey Questions:", len(questions))
print("\nIdentified Factors:")
for factor, name in factor_names.items():
```

## OUTPUT

```
EMPLOYEE SATISFACTION DATASET
   Salary Satisfaction  Benefits  Pay Fairness  Workload  Work Life Balance
0                    4         3             3         3                  3
1                    4         4             4         3                  3
2                    5         4             5         4                  4
3                    3         4             3         2                  3
4                    2         3             3         1                  2

   Job Stress  Manager Support  Career Growth  Team Spirit  Recognition
0           3                3              4            4            4
1           3                3              3            3            3
2           3                3              3            4            3
3           2                2              3            3            2
4           1                1              1            1            1

Dataset Shape: (300, 10)

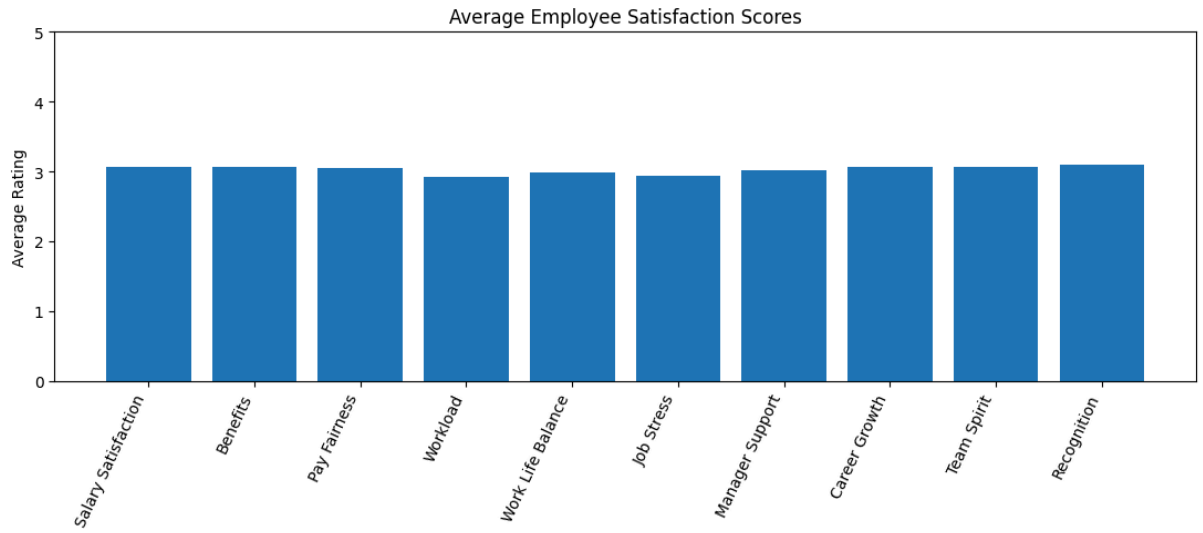
Missing Values: 0

DESCRIPTIVE STATISTICS
       Salary Satisfaction  Benefits  Pay Fairness  Workload  \
count               300.00    300.00        300.00    300.00
mean                  3.06      3.07          3.05      2.92
std                   0.93      0.91          0.91      0.94
min                   1.00      1.00          1.00      1.00
25%                   2.00      2.75          2.00      2.00
50%                   3.00      3.00          3.00      3.00
75%                   4.00      4.00          4.00      4.00
max                   5.00      5.00          5.00      5.00

       Work Life Balance  Job Stress  Manager Support  Career Growth  \
count             300.00      300.00           300.00         300.00
mean                2.98        2.94             3.02           3.06
std                 1.01        0.95             0.98           0.99
min                 1.00        1.00             1.00           1.00
25%                 2.00        2.00             2.00           2.00
50%                 3.00        3.00             3.00           3.00
75%                 4.00        4.00             4.00           4.00
max                 5.00        5.00             5.00           5.00

       Team Spirit  Recognition
count       300.00       300.00
mean          3.06         3.09
std           0.95         0.94
min           1.00         1.00
25%           2.00         3.00
50%           3.00         3.00
75%           4.00         4.00
max           5.00         5.00

AVERAGE SATISFACTION SCORES
Salary Satisfaction    3.06
Benefits               3.07
Pay Fairness           3.05
Workload               2.92
Work Life Balance      2.98
Job Stress             2.94
Manager Support        3.02
Career Growth          3.06
Team Spirit            3.06
Recognition            3.09
dtype: float64
```

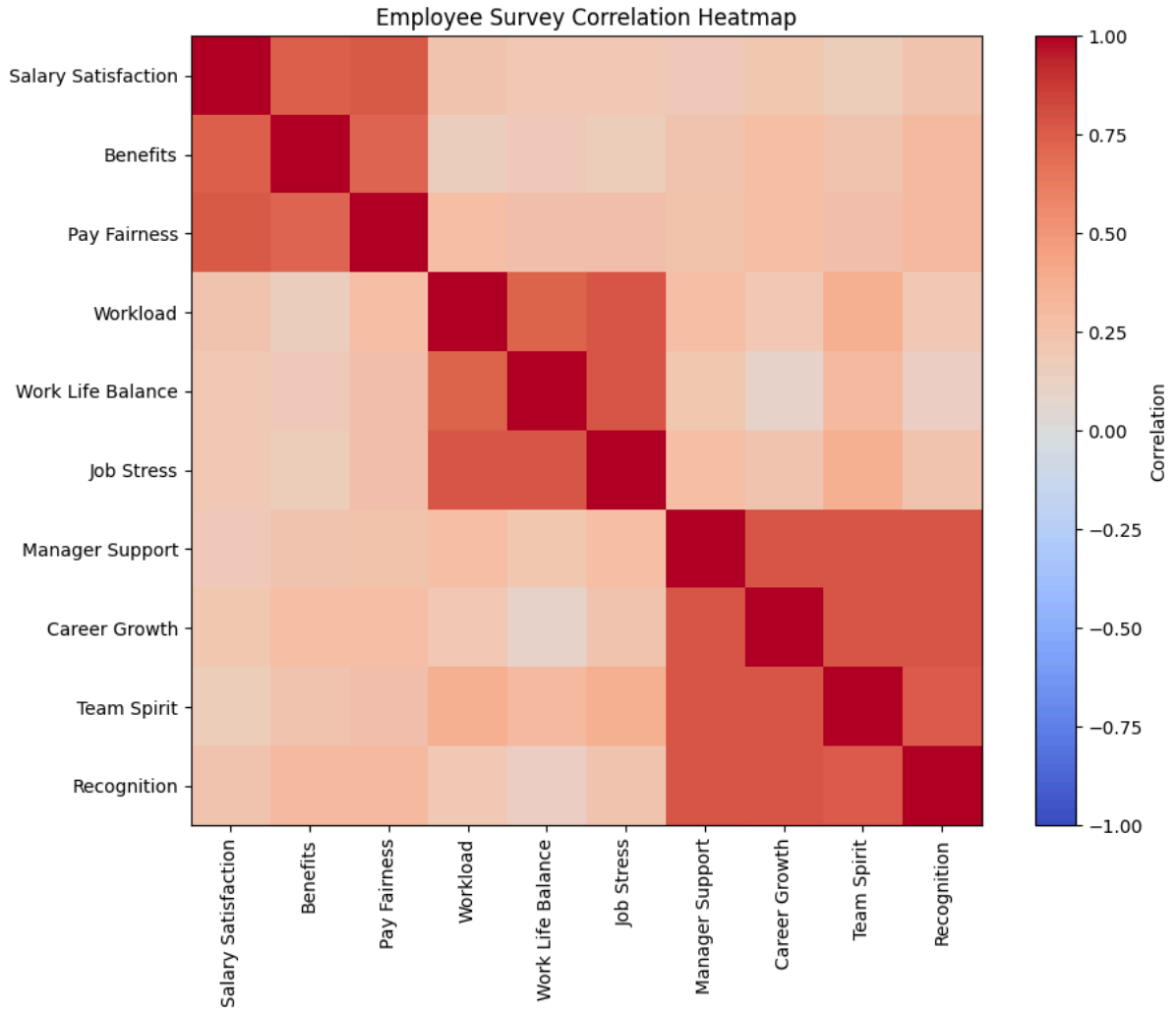


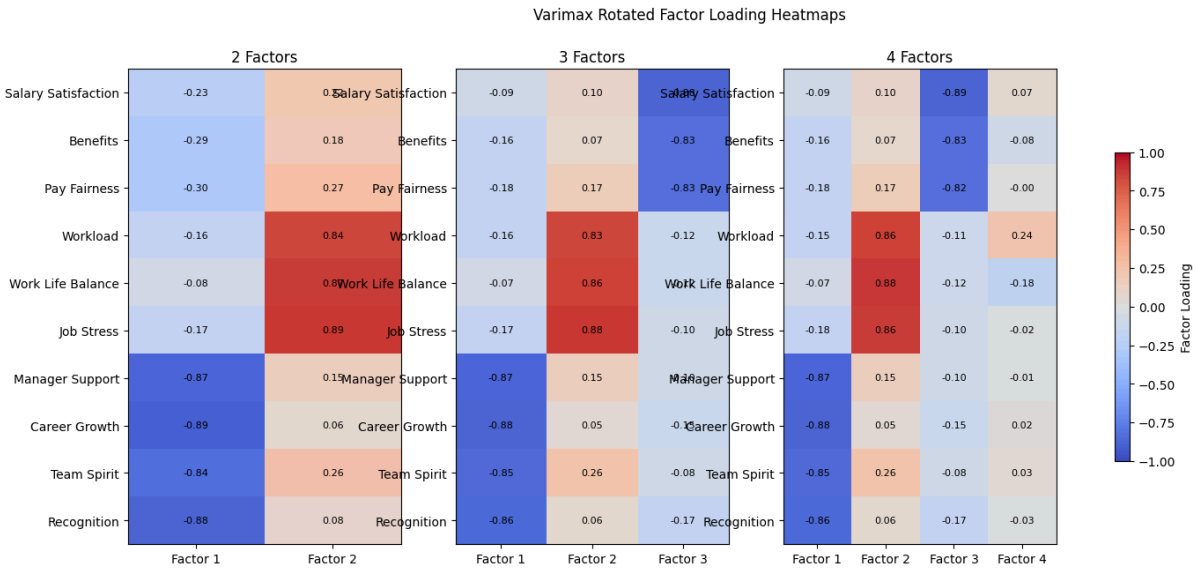

CORRELATION MATRIX

| | Salary Satisfaction | Benefits | Pay Fairness | Workload |
|---|---|---|---|---|
| Salary Satisfaction | 1.00 | 0.75 | 0.76 | 0.22 |
| Benefits | 0.75 | 1.00 | 0.72 | 0.16 |
| Pay Fairness | 0.76 | 0.72 | 1.00 | 0.27 |
| Workload | 0.22 | 0.16 | 0.27 | 1.00 |
| Work Life Balance | 0.19 | 0.18 | 0.26 | 0.73 |
| Job Stress | 0.19 | 0.18 | 0.26 | 0.78 |
| Manager Support | 0.19 | 0.23 | 0.25 | 0.27 |
| Career Growth | 0.21 | 0.27 | 0.29 | 0.19 |
| Team Spirit | 0.17 | 0.22 | 0.26 | 0.37 |
| Recognition | 0.23 | 0.29 | 0.31 | 0.20 |

| | Work Life Balance | Job Stress | Manager Support |
|---|---|---|---|
| Salary Satisfaction | 0.19 | 0.19 | 0.19 |
| Benefits | 0.18 | 0.18 | 0.23 |
| Pay Fairness | 0.26 | 0.26 | 0.25 |
| Workload | 0.73 | 0.78 | 0.27 |
| Work Life Balance | 1.00 | 0.79 | 0.21 |
| Job Stress | 0.79 | 1.00 | 0.28 |
| Manager Support | 0.21 | 0.28 | 1.00 |
| Career Growth | 0.11 | 0.22 | 0.78 |
| Team Spirit | 0.29 | 0.37 | 0.78 |
| Recognition | 0.14 | 0.22 | 0.79 |

| | Career Growth | Team Spirit | Recognition |
|---|---|---|---|
| Salary Satisfaction | 0.21 | 0.17 | 0.23 |
| Benefits | 0.27 | 0.22 | 0.29 |
| Pay Fairness | 0.29 | 0.26 | 0.31 |
| Workload | 0.19 | 0.37 | 0.20 |
| Work Life Balance | 0.11 | 0.29 | 0.14 |
| Job Stress | 0.22 | 0.37 | 0.22 |
| Manager Support | 0.78 | 0.78 | 0.79 |
| Career Growth | 1.00 | 0.78 | 0.78 |
| Team Spirit | 0.78 | 1.00 | 0.75 |
| Recognition | 0.78 | 0.75 | 1.00 |


Employee Survey Correlation Heatmap
Salary Satisfaction
Benefits
Pay Fairness
Workload
Work Life Balance
Job Stress
Manager Support
Career Growth
Team Spirit
Recognition
Correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Varimax Rotated Factor Loading Heatmaps
2 Factors
3 Factors
4 Factors
Factor 1
Factor 2
Factor 3
Factor 4
Factor Loading
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

```
Salary Satisfaction    -0.091     0.103    -0.879
Benefits               -0.163     0.072    -0.825
Pay Fairness           -0.175     0.171    -0.826
Workload               -0.156     0.832    -0.118
Work Life Balance      -0.071     0.858    -0.119
Job Stress             -0.173     0.885    -0.098
Manager Support        -0.871     0.149    -0.097
Career Growth          -0.880     0.050    -0.146
Team Spirit            -0.847     0.256    -0.081
Recognition            -0.861     0.061    -0.174

HIGH LOADING QUESTIONS

Factor 1:
- Career Growth
- Manager Support
- Recognition

Factor 2:
- Job Stress
- Work Life Balance
- Workload

Factor 3:
- Salary Satisfaction
- Pay Fairness
- Benefits

INTERPRETED LATENT FACTORS
Factor 1 : Career Support
Factor 2 : Work Culture
Factor 3 : Compensation

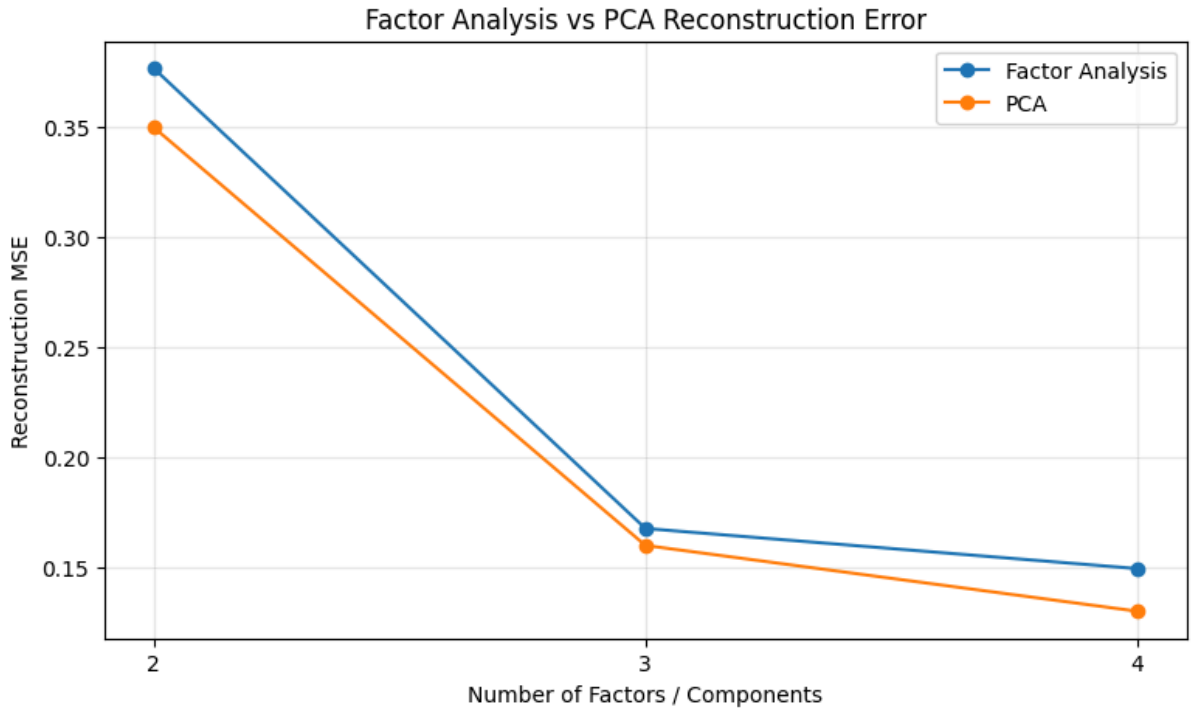
COMMUNALITY AND UNIQUENESS
              Question  Communality  Uniqueness
0  Salary Satisfaction        0.792       0.209
1             Benefits        0.712       0.287
2         Pay Fairness        0.743       0.257
3             Workload        0.730       0.270
4    Work Life Balance        0.755       0.245
5           Job Stress        0.822       0.178
6      Manager Support        0.790       0.210
7        Career Growth        0.797       0.203
8          Team Spirit        0.790       0.210
9          Recognition        0.775       0.225

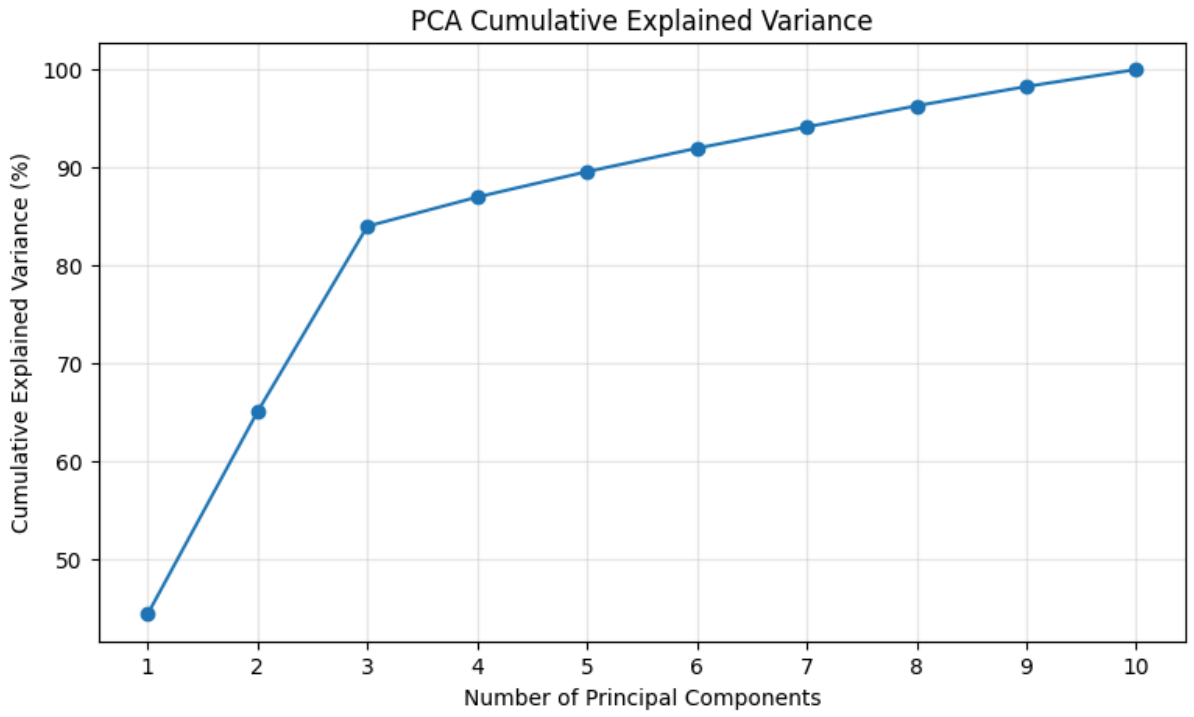
PCA WITH 2 COMPONENTS
Explained Variance: 65.01 %
Reconstruction MSE: 0.3499

PCA WITH 3 COMPONENTS
Explained Variance: 83.97 %
Reconstruction MSE: 0.1603

PCA WITH 4 COMPONENTS
Explained Variance: 86.96 %
Reconstruction MSE: 0.1304

FA VS PCA COMPARISON
         Method  Components  Reconstruction MSE
Factor Analysis           2              0.3767
            PCA           2              0.3499
Factor Analysis           3              0.1680
            PCA           3              0.1603
Factor Analysis           4              0.1497
```




Employees: 300
Survey Questions: 10

Identified Factors:
Factor 1 : Career Support
Factor 2 : Work Culture
Factor 3 : Compensation

## 5. Result

Factor Analysis was implemented using Scikit-learn to identify latent factors influencing employee satisfaction from a simulated dataset of 300 employees and 10 survey questions. Factor models with 2, 3, and 4 factors were fitted, and Varimax rotation was applied to improve interpretability. Correlation matrices, factor loading heatmaps, communalities, and uniqueness values were calculated. The factor-loading patterns were used to interpret underlying dimensions such as Compensation, Work Culture, and Career Support. Finally, Factor Analysis was compared with PCA to demonstrate the difference between identifying hidden factors and reducing dimensionality.